LASITH HAPUTHANTHRI

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PERSONAL STATEMENT

Undergraduate Electronics Engineer with experience in embedded programming, RF electronics, and digital hardware design. Skilled in microcontrollers, embedded C/C++, circuit analysis, and hardware debugging. Familiar with sensors, communication interfaces, and RF concepts. Strong foundation in Electronics Engineering principles.

Interested Area: RF Electronics, PCB Design, Embedded Systems, IoT, Electronic Product Design, Digital System Design, Processor Design, Hardware Acceleration, FPGA, Robotics

EXPERIENCE

Internship Magicbits PVT LTD *Dec 2024 - June 2025*
Employee progress tracking system using HMI devices for Ansell Lanka.

Aura Digital Labs- Electronic Engineer *Since Dec 2024*

Photomath
Math expert *Dec 2021 - Feb 2024*
Math Reviewer *Aug 2022 - Feb 2024*

Brookfab 3D printing- Co-Founder

EDUCATION

University of Moratuwa
B.Sc. Eng. (Hons.) Electronic & Telecommunication Engineering *Aug 2022 - Present*

Dharmaraja College Kandy *2010-2020*

- GCE Advanced Level: Physical science stream, 3A passes for Chemistry, Physics, and Combined Mathematics with a z-score of 2.23 (Island Rank 329)
- GCE Ordinary Level: 9A

PROJECTS

Internship Projects *Dec 2024 - June 2025*

Employee Progress Tracking System for Ansell Lanka *Embedded Systems*

- Tools & Technologies used: **PIstudio, RFID, Modbus Communication, Thingsboard**
- A System can track each employee's work and the progress of each department.
- It used **HMI** devices with **RFID** sensors to collect employee data and visualize it in the ThingsBoard dashboard.
- Implemented the firmware for the HMI and tested in a **Pilot Project with 20 devices** in the field.

University Projects

UAS Detection using CW and FMCW Radar *RF*
Final Year Project *Ongoing*

- Tools & technologies used: HFSS, Altium Designer, MATLAB, VS Code, LT Spice, mmWave Software

- Designing a CW radar system from scratch for drone detection, including RF front-end and signal processing. Integrated an FMCW radar evaluation board for range and velocity estimation, focusing on practical radar detection techniques and real-world interference handling.
- The project includes two **Teflon PCBs** transmitter and receiver antennas, and a **6-layer PCB** for CW signal the Processing.
- Contributing to RF antenna designing, PCB designing, and CW signal processing.

Echo-watt: Inverter Power Monitoring Device *Embedded Systems*
EN4440: Emmbeded System Enginnering *2025*

- Implemented a microcontroller-based embedded system that polls inverter parameters, buffers and compresses data, and uploads securely to the cloud on a constrained schedule. The system supports remote configuration, fault handling, and firmware-over-the-air (FOTA) updates.
- Contributed to designing the Architecture, Modbus polling, MQTT structure, and the Dashboard.

Serial Bus Design *FPGA*
EN4021: Advanced Digital Systems *2025*

- Tools & technologies used: Quartus Prime
- Designed and implemented a wired Bus architecture for two masters and three slaves with split transactions and support for dual-split. The system is demonstrated on the DE0-Nano FPGA board.

SIMD Processor Design for CNN-based ML models *FPGA*
EN4021: Advanced Digital Systems *2025*

- Tools & technologies used: Xilinx Vivado and Vitis
- Designed a SIMD processor architecture to accelerate CNN computations. Contributed to developing the modules for polling and testing the systolic arrays.

Single Cycle Processor *FPGA*
EN3021 Digital System Design Module *2025*

- 32-bit pipelined single-cycle RISC-V processor, implementing core modules like the Program Counter, Instruction Register, and Multiplexers in SystemVerilog. A test bench is being developed to verify functionality and ensure compliance with RISC-V standards.

Floating point Adder, Subtractor, Multiplier, and Divider for IEEE 754 standards *FPGA*
EN3021 Digital System Design Module *2025*

- Tools & technologies used: Quartus Prime
- Developed a 32-bit floating-point Adder, Subtractor, Multiplier, and Divider following IEEE 754 standards with a testbench to validate the accuracy and performance of all operations. The system is tested on the DE2-70 FPGA board.

LUNA: Vision-based Automated Restaurant Robot *Robotics*
EN2160 Electronic Design Realization Module Project *2024*

- Developed a restaurant robot using ROS and Kinect2, equipped with a Kalman-filter-based food stabilization tray to avoid beverage spillage.
- Contributed to Hardware and Firmware development, PCB designing, and implemented the Kalman filter-based stabilizing tray.

External Projects

SteerSafe -EOG-Based Drowsiness Detection Wearable Spectacles *RF, Embedded Systems*
IEEE Circuit Challenge 2024 Champion, Brainstorm 2024 1st Runners up, *2024*

- Smart Spectacle frame and smartwatch that collect EOG data to predict driver's sleepiness using Machine Learning and alert the driver before an accident occurs.
- The spectacles and smartwatch communicate with the Raspberry Pi using NRF.

- The design includes Hi-density **6-layer** and **4-layer** PCBs with **Inverted F antenna**.

HydroLink 2.0 - Smart Water Tank Monitoring and controlling System *Embedded Systems*
UBLC World Bank Grant *2024*

- An IoT device plugged into any domestic water tank checks the water level and sends it to Firebase. The water level can be monitored via a custom app, allowing users to control it directly from the app.
- UBLC World Bank funding to produce and sell 10 devices, supporting the initiative's growth and market presence.

REPLACE: Smart and Intelligent Wall Socket *Embedded Systems*
IEEE Challenge Sphere Arduino Challenge *2024*

- A smart wall socket that can detect anomalies in the domestic power grid. With a mobile app (React Native), users can monitor power consumption and costs and control the socket.

Vision-based Myocardial Infarction Detection System *Machine Learning, Software*
Personal Interest *2025*

- A system that includes a mobile app (Flutter) and a Python backend to detect Myocardial Infarction (Heart attack) using Echocardiogram videos.
- Have used a CNN + BiLSTM model to predict the patient's MI probability.

TECHNICAL SKILLS

Languages Sinhala(Native), English
Programming Python, C, C++, Matlab, SystemVerilog, Verilog, JavaScript, Lua
Soft Skills Problem Solving, Strategic Decision-Making, Project Management, Team Leadership
Developer Tools: Altium Designer, Solidworks, Figma, MATLAB antenna toolbox, Simulink, Jupyter, Multisim, LTspice, Latex, PIStudio, Ansys HFSS, Spring Boot

HONORS AND AWARDS

- **Champion (National Rank)| Silicon Pulse** *2025*
  A competition for designing and implementing the given task using only analog components without using outside resources.
- **Champion (National Rank)| IEEE Challenge Sphere Arduino Challenge** *2024*
  REPLACE: A bright and Intelligent system that is designed with custom-made analog sensors and made to be scalable and for anomaly detection.
- **Champion (National Rank) | IEEE Challenge Sphere Circuit Challenge** *2024*
  StearSafe: A spectacle frame and watch that can use the EOG signals from the user to measure drowsiness and alert the user.
- **1st Runners up (National Rank) | Brainstorm The Premier Biomedical Competition** *2024*
  StearSafe: A spectacle frame and watch that can use the EOG signals from the user to measure drowsiness and alert the user.
- **Champion (National Rank)| SLIOT Challenge 2023- University Category** *2023*
  HydroLink 2.0: An IoT device that can turn any domestic water tanks into smart ones.
- **Special Award for Chess**
  Dharmaraja College Kandy *2013,2016*
  Sri Lankan representation for World Youth and Asian Youth Chess Championships 2013 and got selected to World Youth and Asian Youth Chess Championships 2016.

EXTRA-CURRICULAR ACTIVITIES

Volunteering

- **Sasnaka Sansada Project Ganitha Saviya** *2021-Present*
  Sasnaka Sansada Ganitha Saviya is a Sri Lankan initiative that promotes and enhances mathematical education and problem-solving skills among high school students through competitions, workshops, and other educational activities.
- **Sri Lanka Robotics Challenge**
  - Organizing Committee *2024,2025*
  - School Workshops Presenter *2024*
  - Projects Demonstration, LUNA: Vision-based Automated Restaurant Robot *2024*
- **EXMO 2023 Technological Exhibition** *2023*
  EXMO 2023 is the University of Moratuwa's flagship technology exhibition in Sri Lanka.
  - Digital Lab Coordinator
  - Projects Demonstration
    HydroLink - Smart water tank monitoring and controlling Device.

Sports and Games

- **Champion** in Sri Lanka All-island Youth Chess Championship 2012.
- **Champion** from team section in 4th Asian School Chess Championship 2008.
- **2nd Runners up** from the individual section in the 4th Asian School Chess Championship 2008, team section in the Youth Asian Chess Championship 2009. Selected to the World School Chess Championship 2016, 3-11 December in Sochi, Russia.
- Selected to 10th Asian School Chess Championship 2016 from 9-18 2016 in Tehran, **Iran** at U.12 2013 World Youth Chess Championships, Al-ain, **UAE**, at U.12 2013 Asian Youth Chess Championships, Mazandaran, **Iran**, 19-27 June 2013, Sri Lanka at the 8th Asian Schools Chess Championship 2012, New Delhi, **India** 13-20 December 2012, Sri Lanka at U.8 2009 Asian Youth Chess Championships, New Delhi, **India**, Sri Lanka at the 6th Asian Schools Chess Championship 2010, Colombo, Sri Lanka, Sri Lanka at the 5th Asian Schools Chess Championship 2009, Colombo, Sri Lanka, Sri Lanka at the 4th Asian Schools Chess Championship 2008, Colombo, Sri Lanka.
- Member of the University Table Tennis pool.

Club and Organizations

- Member of the Electronic Club of University of Moratuwa *2022-Present*
- IEEE Student Member *June 2024 -Present*
- Chess Team, Dharmaraja College, Kandy, Sri Lanka. *2009-2020*
  - Captain *2013-2019*
  - Vice-Captain *2010-2012*

REFERENCES

- **Dr. Chamira U. S. Edussooriya**
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